TOSHIBA GTR Module Silicon N Channel IGBT

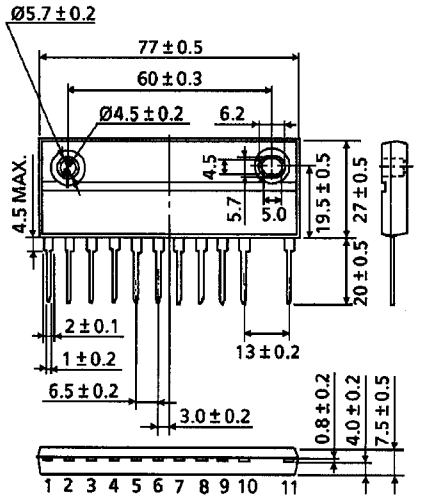
MP6750

High Power Switching Applications

Motor Control Applications

- The electrodes are isolated from case.
- 6 IGBTs are built into 1 package.
- Enhancement-mode
- Low saturation voltage
  : $V_{CE\ (sat)}$ = 4.0V (Max) (I_C = 15A)
- High speed : t_f = 0.35μs (Max) (I_C = 15A)
  t_{rr} = 0.15μs (Max) (I_F = 15A)

Unit: mm

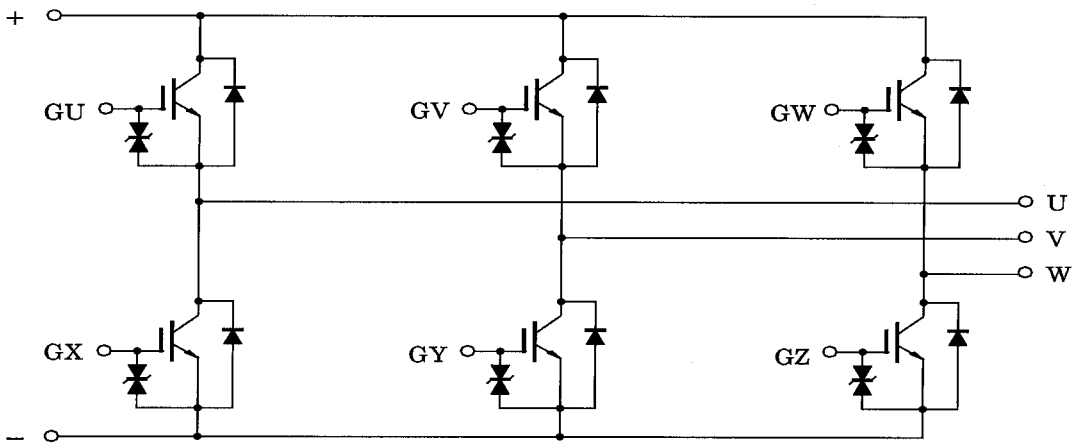


1. (+)	2. GU	3. U	4. GV	5. V	6. GW
7. W	8. GX	9. GY	10. GZ	11. (−)	

JEDEC	—
EIAJ	—
TOSHIBA	2-78A1A

Weight: 44g

Equivalent Circuit



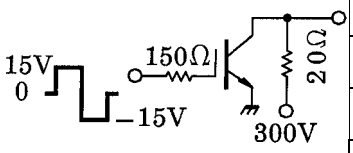
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Maximum Ratings (Ta = 25°C)

Characteristic		Symbol	Rating	Unit
Collector-emitter voltage		V_{CES}	600	V
Gate-emitter voltage		V_{GES}	± 20	V
Collector current	DC	I_C	15	A
	1ms	I_{CP}	30	
Forward current	DC	I_F	15	A
	1ms	I_{FM}	30	
Collector power dissipation (Tc = 25°C)		P_C	55	W
Junction temperature		T_j	150	°C
Storage temperature range		T_{stg}	−40 ~ 125	°C
Isolation voltage		V_{Isol}	2500 (AC 1 minute)	V
Screw torque		—	1.5	N·m

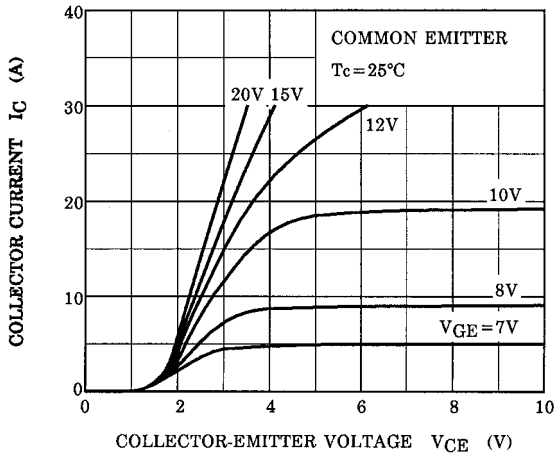
Electrical Characteristics (Ta = 25°C)

Characteristic		Symbol	Test Condition	Min	Typ.	Max	Unit
Gate leakage current		I_{GES}	V_{GE} = ± 20V, V_{CE} = 0	—	—	± 20	μA
Collector cut-off current		I_{CES}	V_{CE} = 600V, V_{GE} = 0	—	—	1.0	mA
Gate-emitter cut-off voltage		$V_{GE\ (off)}$	I_C = 15mA, V_{CE} = 5V	3.0	—	6.0	V
Collector-emitter saturation voltage		$V_{CE\ (sat)}$	I_C = 15A, V_{GE} = 15V	—	3.0	4.0	V
Input capacitance		C_{ies}	V_{CE} = 10V, V_{GE} = 0, f = 1MHz	—	1000	—	pF
Switching time	Rise time	t_r		—	0.3	0.6	μs
	Turn-on time	t_{on}		—	0.4	0.8	
	Fall time	t_f		—	0.2	0.35	
	Turn-off time	t_{off}		—	0.5	1.0	
Forward voltage		V_F	I_F = 15A, V_{GE} = 0	—	1.7	2.5	V
Reverse recovery time		t_{rr}	I_F = 15A, V_{GE} = −10V di / dt = 50A / μs	—	0.08	0.15	μs
Thermal resistance		$R_{th\ (j\text{-}c)}$	Transistor	—	—	2.27	°C / W
			Diode	—	—	3.09	

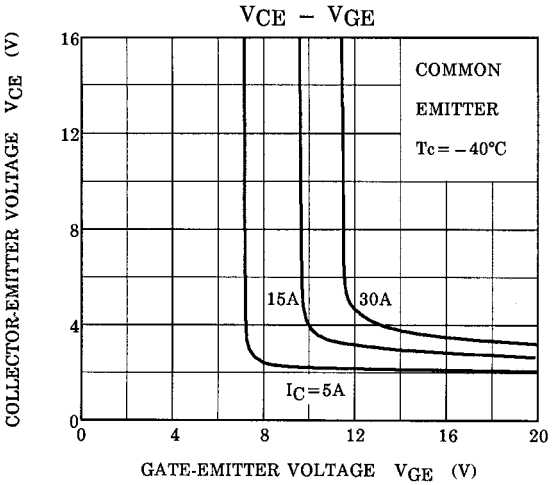
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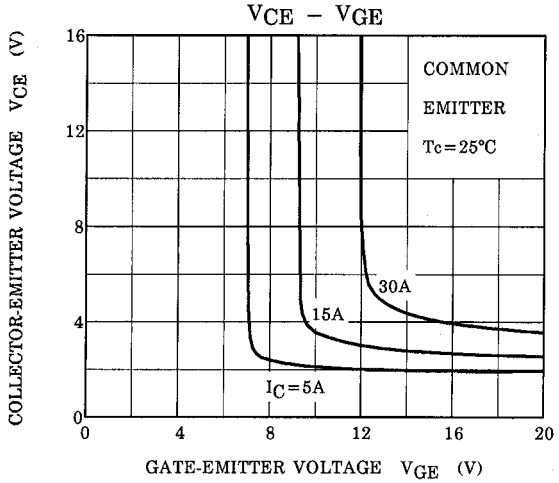
I_C – V_{CE}



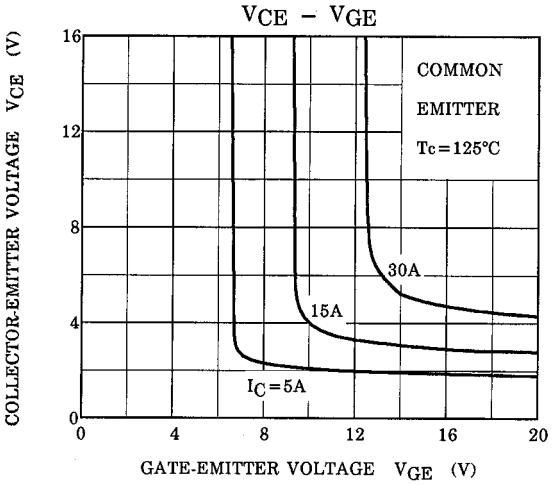
V_{CE} – V_{GE}



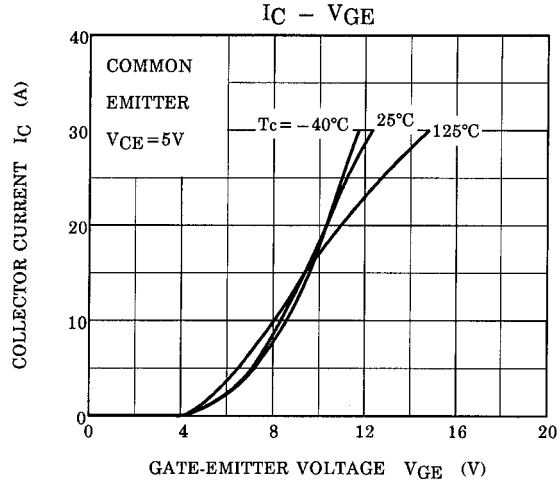
V_{CE} – V_{GE}



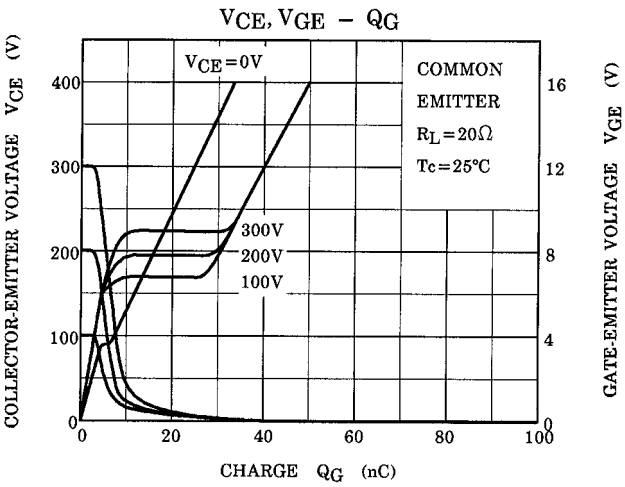
V_{CE} – V_{GE}



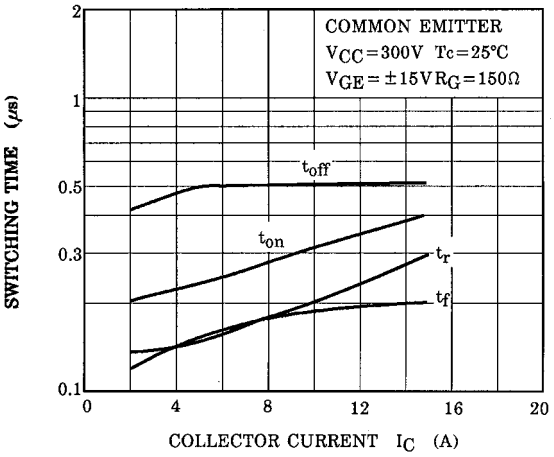
I_C – V_{GE}



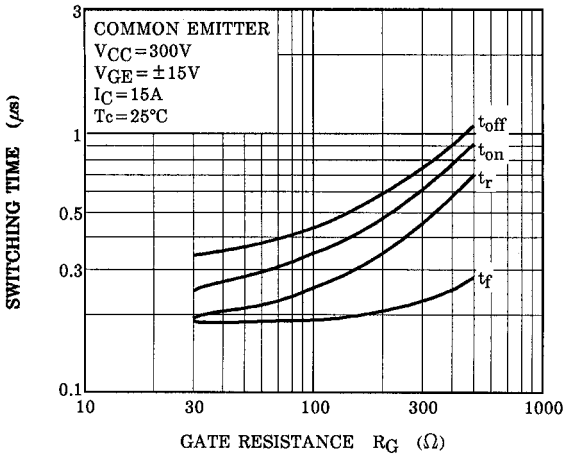
V_{CE}, V_{GE} – Q_G



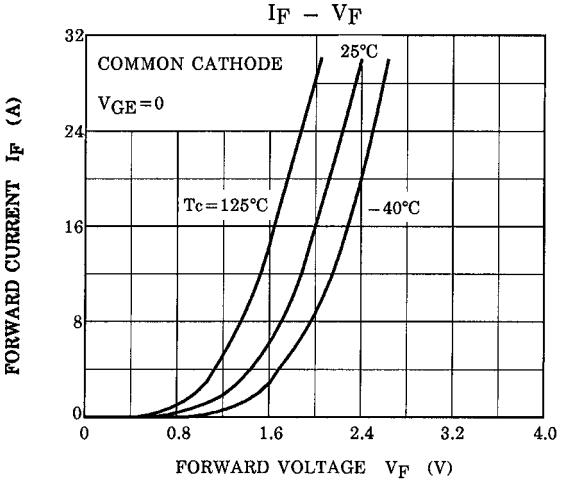
SWITCHING TIME – I_C



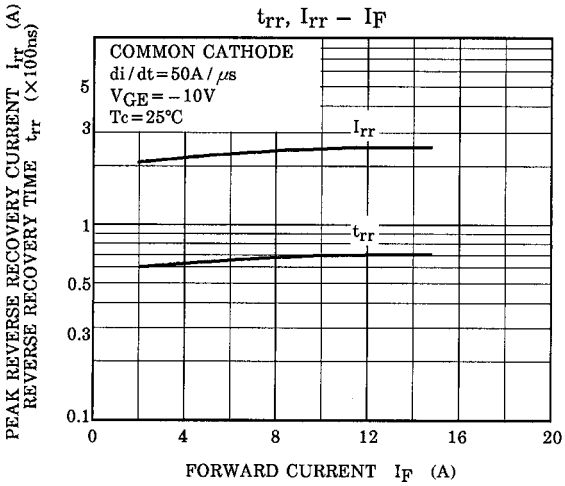
SWITCHING TIME – R_G



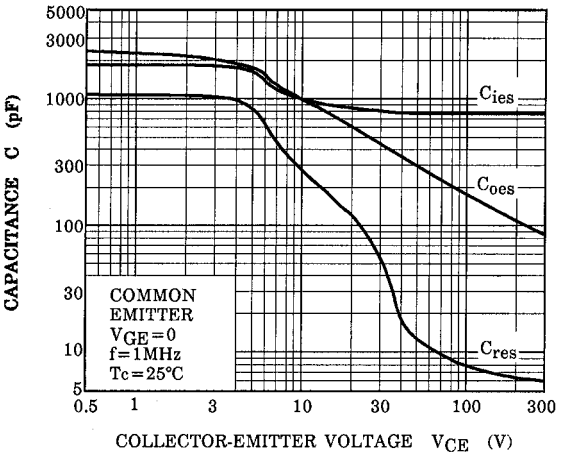
I_F – V_F



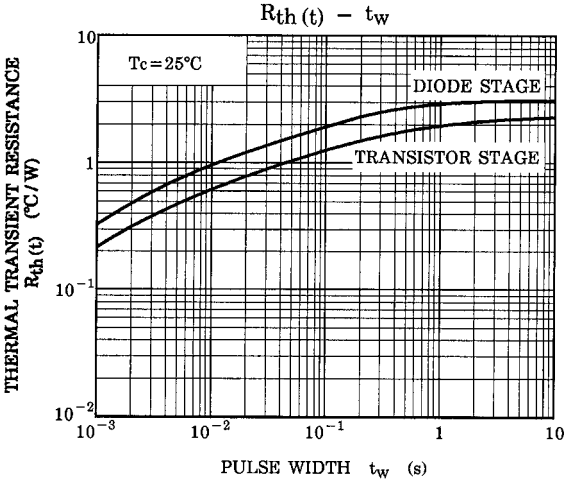
t_{rr}, I_{rr} – I_F



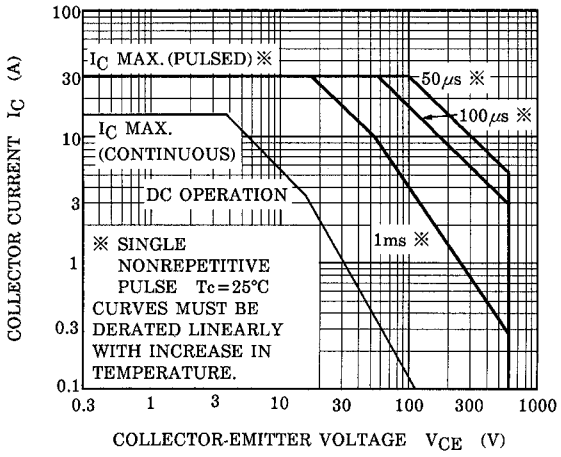
C – V_{CE}



R_{th} (t) – t_w



SAFE OPERATING AREA



REVERSE BIAS SOA

